POWER BI

Power BI Interview Notes

The model, the DAX and the concepts that come up in every Power BI round.

- **Star schema, explained simply**
- **6 DAX measures you will reuse**
- **Top questions with answers**

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Model it right

Most slow or wrong reports are a modelling problem, not a DAX problem.

- **Star schema**

  One fact table (Sales) in the middle. Dimension tables (Date, Product, Customer) around it.

- **One-to-many, single direction**

  Filters flow from dimension to fact. Avoid both-direction filters unless you must.

- **Always add a Date table**

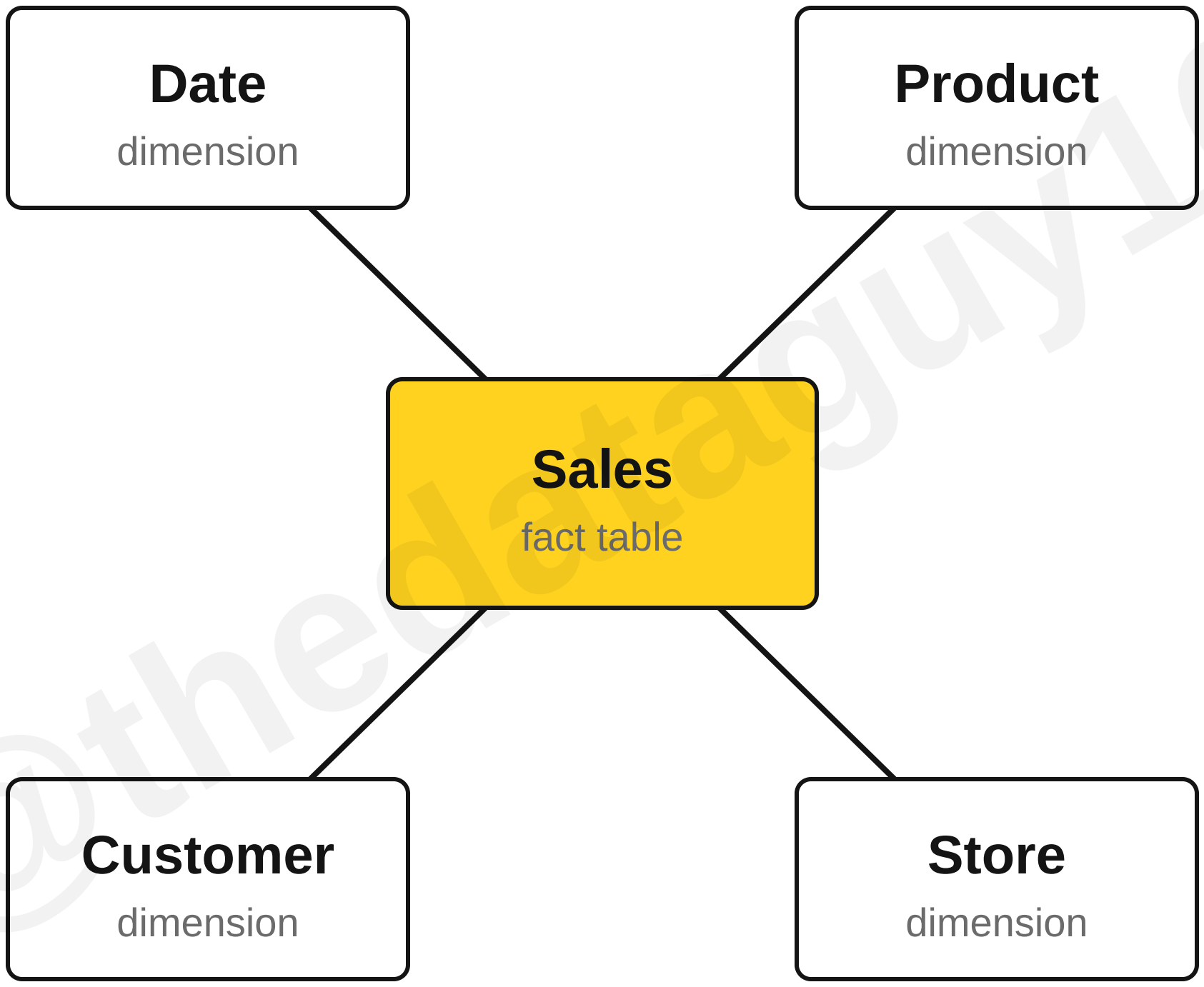
  Mark it as a date table. Time intelligence functions need it.

```
Date = CALENDAR(DATE(2024,1,1),
DATE(2026,12,31))
```

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Star schema at a glance

Draw this in the interview. It shows you understand modelling, not just visuals.



Each line: 1 → many. Filters flow into the fact table.

Rule: numbers you add up live in the fact table. Things you filter or group by live in dimensions.

6 measures you will reuse

Write these once and you can build most business dashboards.

```
Total = SUM(Sales[Amount])

North = CALCULATE([Total],
Region[Name]="North")

% of Total = DIVIDE([Total],
    CALCULATE([Total], ALL(Region)))

Sales LY = CALCULATE([Total],
   SAMEPERIODLASTYEAR('Date'[Date]))

YoY % = DIVIDE([Total] - [Sales LY],
[Sales LY])

YTD = TOTALYTD([Total], 'Date'[Date])
```

DIVIDE instead of "/" returns blank instead of an error when the bottom is 0.

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Power Query: clean before you model

Do the cleaning here, not in DAX. Every step is recorded and re-runs on refresh.

Change type

Set dates, whole numbers and text first. Wrong types break relationships.

Remove duplicates and blanks

Right-click column → Remove Duplicates. Filter out nulls.

Unpivot

Turn Jan, Feb, Mar columns into Month and Value rows. Needed for time analysis.

Merge vs Append

Merge = JOIN (adds columns). Append = UNION (adds rows).

Group By

Pre-aggregate huge tables to make reports faster.

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Concepts they test

Know the one line answer, then give an example.

Measure vs calculated column

Column: computed row by row at refresh, stored, used for slicing. Measure: computed at query time, used for totals and KPIs.

Row context vs filter context

Row context is the current row. Filter context is what slicers, visuals and CALCULATE have filtered.

SUM vs SUMX

SUMX calculates row by row first, then adds. Use it for Qty × Price.

```
SUMX(Sales, Sales[Qty] * Sales[Price])
```

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Most asked questions

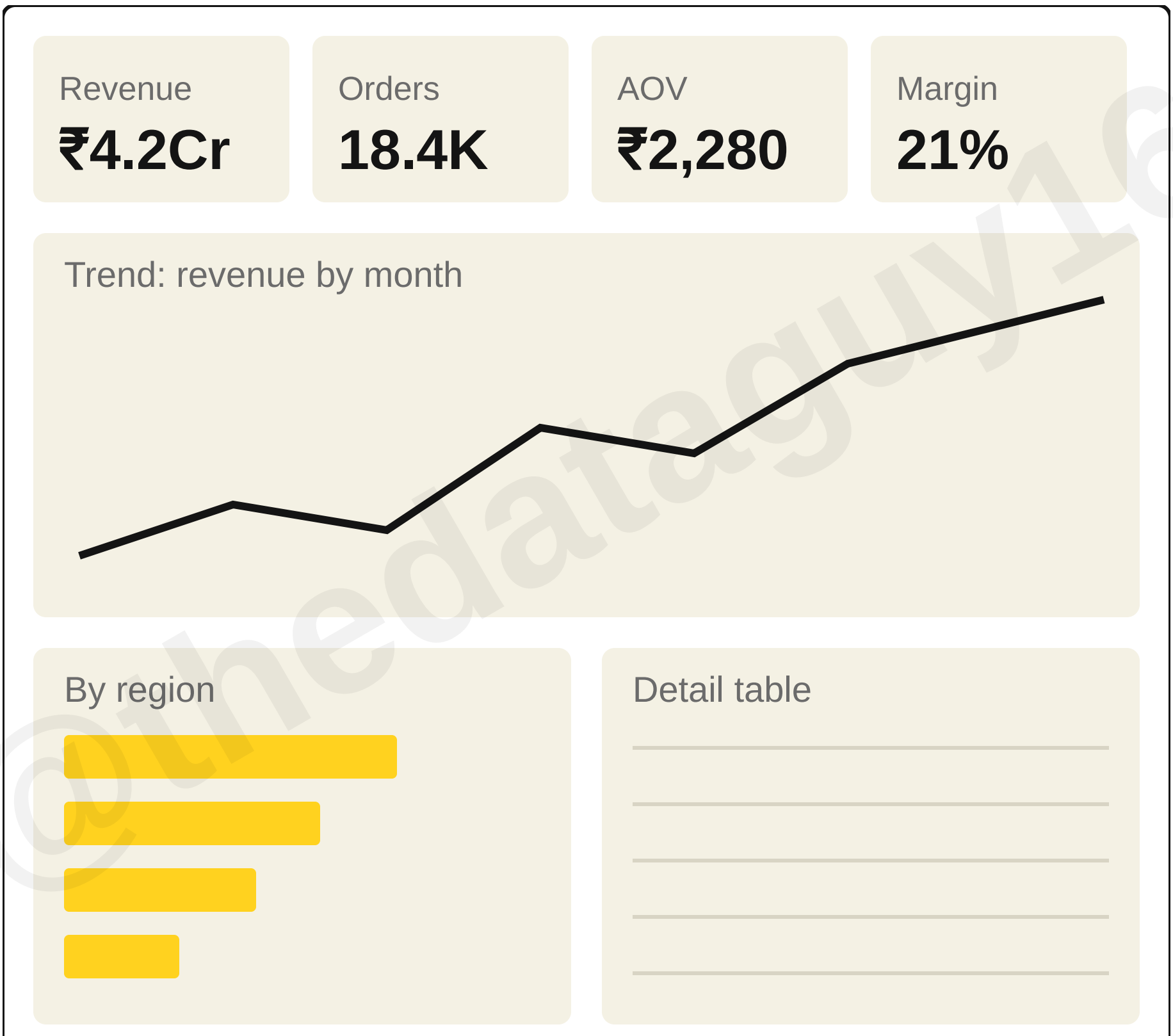
Prepare a 2 line answer for each.

- **What does CALCULATE do?**
  Changes the filter context, then evaluates the expression.
- **Import vs DirectQuery**
  Import is fast, data is copied. DirectQuery is live, slower, for very large or real-time data.
- **Row Level Security**
  Roles with DAX filters so each manager sees only their region.
- **Slow report?**
  Fewer visuals per page, remove unused columns, measures over calculated columns, star schema.

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Dashboard layout that works

Managers read top-left to bottom-right. Put the answer first, detail last.



Top row

3-4 KPI cards with comparison to last period.

Middle

One trend chart. Bottom: breakdown and detail table.